

Speaker notes v2.0 — medium

Graph-Based vs Transition-Based Dependency Parsing for Uzbek

EDM 2026 (≈ 7.5 min, relaxed pace)

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How to use. One short paragraph per slide — enough to keep the “why,” but trimmed so a calm delivery still fits 7.5 minutes. Bold phrases are the ones to land; **[pause]** is a deliberate beat.

Three numbers to land: LAS 63.81 vs 47.11 (a 16.70 gap) and cross-treebank +9.62 LAS.

Slow down on: Slide 7 (fusion) and Slide 10 (the gap). Everything else can move briskly.

If you finish early, pause — don't pad with detail.

Slide 1 — Title

[0:00 → 0:20 min]

Good morning, and thank you to the organisers. I am **Sanatbek Matlatipov**, from the National University of Uzbekistan, with co-authors Maftuna Ismoilova and Mersaid Aripov. Our question is simple but unanswered for Uzbek: when data is scarce, **which parser should you reach for — graph-based or transition-based — and why?**

Slide 2 — Why Uzbek is hard

[0:20 → 1:00 min]

Two properties make this hard. Uzbek is **agglutinative** — a word stacks suffixes for case, tense, person — and it is **low-resource**, with only 1,184 UD sentences total. The box on the right is the key idea: take *bolalarning*, “of the children” — stem, plural, and the genitive **-ning**. That final suffix *is* the relation. So in Uzbek, **miss the suffix and you miss the parse.**

Slide 3 — Contributions

[1:00 → 1:30 min]

Three contributions. First, the **first head-to-head comparison** of a graph-based parser, Stanza, and a transition-based parser, spaCy — both on the **same** BERT, so architecture is the only variable. Second, we show cross-treebank data helps the two **very differently**. Third, an error analysis showing where each one wins. Everything is public.

Slide 4 — The two treebanks

[1:30 → 2:00 min]

We use two treebanks: **UzUDT** — 684 sentences of fiction and academic text — and **UT** — 500 sentences of news and fiction. Neither had a standard split, so we re-split both. Merged, that is 781 training sentences — roughly **double** the data over UzUDT alone.

Slide 5 — Why they are complementary

[2:00 → 2:30 min]

And this is not just *more* data — it is **broad**er data. UT is news-heavy, with far more proper nouns and light-verb compounds; UzUDT is literary, with more adverbial clauses and richer morphology. They fill each other's blind spots — which is why merging works so well later.

Slide 6 — Two architectures

[2:30 → 3:05 min]

Same encoder, two very different decoders. **Stanza** is graph-based: it scores *every* head-dependent pair and decodes the best tree globally. **spaCy** is transition-based: it builds the tree incrementally, left to right. They even fuse BERT sub-words differently — last piece vs. average — and that choice matters, so let me take it next.

Slide 7 — Subword fusion (SLOW DOWN)

[3:05 → 3:50 min]

BERT splits words into **WordPieces**, but UD annotates whole tokens, so we must pool several pieces into one. Look at the table: the *last* piece always carries the decisive suffix — genitive, accusative, converb.

[pause]

So **averaging** all pieces dilutes that signal; **keeping the last piece** preserves it. That one change is worth **+2.6 LAS**, with no extra parameters — and it helps parsing, not tagging.

Slide 8 — Setup

[3:50 → 4:15 min]

Seven models across the two frameworks. We use each one's **documented defaults** — a fair out-of-the-box comparison, the way a practitioner actually meets these tools — and the standard UD metrics: tagging, plus unlabeled and labeled attachment.

Slide 9 — Results per framework

[4:15 → 4:50 min]

Two quick reads. **Inside each framework**, the best system uses **merged data**. And in both, the single biggest jump comes from merging — up to eleven LAS points. That sets up the comparison.

Slide 10 — The comparison (SLOW DOWN)

[4:50 → 5:40 min]

This is the core result — same encoder, same data, **only the architecture differs**. On tagging, spaCy is better. But on **labeled parsing**, Stanza scores **63.81** against **47.11** — a **16.70-point gap**, among the largest reported.

[pause]

And the punchline: Stanza wins *despite* worse tagging. So for Uzbek, the **architecture** — not tagging accuracy — drives parse quality, because exhaustive scoring suits a rich case system.

Slide 11 — Augmentation differs

[5:40 → 6:10 min]

Merging helps both — but **unevenly**. For tagging the gains are equal. For parsing they diverge: Stanza gains **+9.62 LAS**, spaCy only +1.76. The graph parser is the one that actually exploits the broader data; spaCy instead gains most on morphology.

Slide 12 — Error analysis

[6:10 → 6:40 min]

Where do errors cluster? In the **long-distance relations** — coordination and noun modification — while local relations are fine. These structurally hard cases are exactly where exhaustive graph scoring has the advantage.

Slide 13 — Examples

[6:40 → 7:00 min]

Two real outputs. On the left, a **perfect parse**, including the tricky genitive chain and a converb. On the right, a **failure** on a light-verb construction — gold in blue, the wrong arcs in red. Rare patterns like this are what a small treebank under-covers.

Slide 14 — Conclusion

[7:00 → 7:25 min]

To conclude: a clear trade-off — graph-based parses far better, **+16.70 LAS** — and cross-treebank data is the **biggest lever**. So our recommendation for Uzbek is concrete: **graph-based parser, monolingual BERT, last-subword fusion**. Next steps: multi-seed testing and broader frameworks.

Slide 15 — Thank you

[7:25 → – min]

Thank you — the data, code, and checkpoints are all released, and I am happy to take your questions.

If asked — short answers

Fair comparison? Defaults for both — and the 16.70 gap is far larger than any tuning effect.

Single runs? A bootstrap CI is about ± 2.4 LAS — well below the gap — and the ranking is consistent across all runs.

Why last-subword? Uzbek suffixes sit at the end, so the last piece carries the syntactic cue. +2.6 LAS, no extra cost.

Size or genre? Both — merging doubles the data *and* the treebanks are genre-complementary.

Link to EDM? Uzbek parsing underpins educational NLP — writing feedback, readability — for 37 million speakers.